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To test this hypothesis, wild-type (WT) *S. typhimurium* cells were transformed with plasmids encoding components of the *lac* operon (Figure 1): β -galactosidase (*lacZ*), lactose permease (*lacY*), transacetylase (*lacA*), or all three (*lac*). Each plasmid also contained genes for the *lac* repressor (*lacI*), which can inhibit *lac* operon transcription in the absence of lactose; the cAMP receptor protein (*crp*), which induces *lac* expression in low glucose; and a chloramphenicol resistance gene (*cat*).

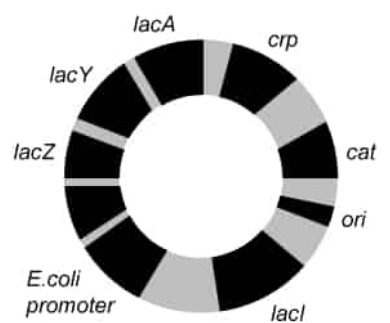


Figure 1 Plasmid pYM-LacZYA (*lac*), used to transform WT *S. typhimurium*. Similar plasmids containing only one of the three *lacZ*, *lacY*, or *lacA* genes were also used.

Virulence assays were performed in mice by inoculating with WT or *lac*⁺ *S. typhimurium*. The assays confirmed that *lac*⁺ bacteria were less virulent than their WT counterparts, and that both WT and *lac*⁺ bacteria were more virulent in the presence of glucose. Subsequently, several genes related to invasion were analyzed for their expression levels in the presence or absence of *lac* plasmids (Table 1; Figure 2).

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Genes		Fold Change(<i>lac</i> ⁺ /WT)	p-value
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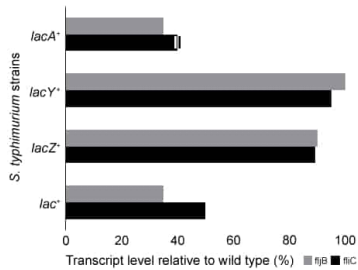
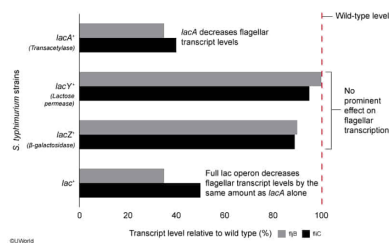


Figure 2 Transcript levels of two *S. typhimurium* flagellar genes, *fliB* and *fliC*, in bacteria transformed with *lacZ*, *lacY*, *lacA*, or *lac* plasmids. Levels are portrayed as a percentage of WT levels.

Based on Figure 2, which of the following would most likely reestablish the virulence of *S. typhimurium* transformed with *lac*?

- ✔ ☒ A. Excising the *lacA* sequence from the plasmid [43%]
- ☐ B. Stimulating transcription of the *lac* operon [15%]
- ☐ C. Removing the mRNA that encodes β -galactosidase [17%]
- ☐ D. Increasing expression of lactose permease [24%]

Explanation:



Expression of **one gene may alter expression of other genes**, which can lead to **altered activity**. Figure 2 shows that expression of the full *lac* operon (ie, the *lac*⁺ strain) in transformed *S. typhimurium* cells resulted in decreased expression of flagellar genes. The passage states that *lac*⁺ *S. typhimurium* cells also experienced a loss of virulence. Therefore, **flagella are likely necessary for these bacteria to be virulent**.

To restore virulence, the specific gene or genes within the operon that altered flagella expression **must be excised**. Figure 2 shows that bacteria transformed with **only the *lacA* gene** also experienced a loss of flagellar gene transcription. In contrast, *lacY* and *lacZ* did not have a significant effect on flagellar genes, so *lacA* alone is most likely responsible for the decrease in virulence. Therefore, **excising this gene** from transformed bacteria will **restore the expression of flagellar genes** and most likely **reestablish virulence**.

(Choice B) Stimulating transcription of the *lac* operon will increase the production of all three *lac* gene products, including the transacetylase encoded by *LacA*. Figure 1 indicates that transacetylase expression *impairs* virulence and would not reestablish it.

(Choice C) β-galactosidase is encoded by *lacZ*. Because *lacZ* does not suppress the transcription of flagellar genes, its removal is unlikely to reestablish virulence.

(Choice D) Lactose permease is encoded by *lacY*. Because addition of *lacY* did not significantly increase the transcription of flagellar genes, it is unlikely to restore flagellar transcription in *lac*⁺ *S. typhimurium*.

Educational objective:

Expression of one gene can affect the expression level of other genes and can alter activities such as virulence. In contrast, different genes may not significantly alter levels of other genes and do not affect their activities. The effect of one gene on others can be tested empirically.

Time spent: 0 sec
Subject:
Biology

QID: 400480
System:
Cellular Biology

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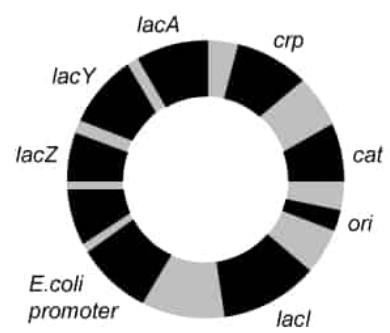


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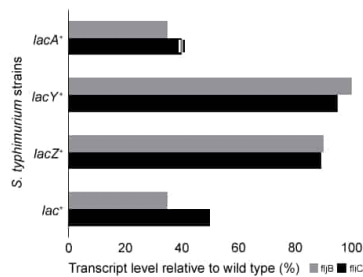
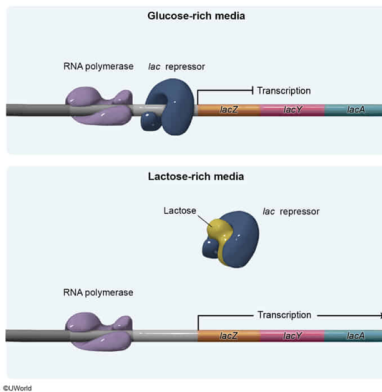


Figure 2 Transcript levels of two *S. typhimurium* flagellar genes, *fliB* and *fliC*, in bacteria transformed with *lacZ*, *lacY*, *lacA*, or *lac* plasmids. Levels are portrayed as a percentage of WT levels.

Which of the following genes would be the most advantageous for *lac*⁺ *S. typhimurium* bacteria in glucose-rich, lactose-poor media?

- ☐ A. *cat* [3%]
- ☒ B. *lacI* [72%]
- ☐ C. *lacZ* [16%]
- ☐ D. *crp* [7%]

Explanation:



Gene expression requires energy in the form of ATP, so the ability to inhibit expression of unnecessary genes is beneficial. Glucose is the preferred carbohydrate fuel of most organisms; when glucose is present in abundance, metabolism of lactose is not necessary. Therefore, for *lac*⁺ *S. typhimurium* **downregulation** of the *lac* genes would be advantageous when in glucose-rich media. The ***lac* repressor**, encoded by *lacI*, binds the operon and inhibits transcription of *lac* genes.

At high lactose concentrations, lactose binds the *lac* repressor and inhibits its interaction with the operon, allowing genes to be transcribed. When glucose is abundant and lactose is low or absent, on the other hand, the *lac* repressor remains unbound and becomes active. Therefore, *lacI* provides a significant advantage for *lac*⁺ bacteria in glucose-rich media because it prevents unnecessary consumption of ATP.

(Choice A) *cat* encodes chloramphenicol resistance. It is beneficial in the presence of chloramphenicol but does not respond to glucose concentration.

(Choice C) *lacZ* encodes β -galactosidase, which is involved in lactose metabolism. It is only beneficial when lactose is readily available and glucose is not. Its expression would be detrimental in glucose-rich conditions, as it would unnecessarily consume ATP.

(Choice D) *crp* is activated by cyclic AMP (cAMP), which is produced in *low glucose* conditions. When activated, *crp* binds the operon and **enhances transcription** of *lac* genes to make better use of lactose as an alternate fuel source. It would therefore be beneficial in low glucose conditions but not in glucose-rich media.

Educational objective:

The *lacI* gene codes for the *lac* repressor and prevents transcription of the *lac* genes. When sufficient glucose is available, the repressor helps conserve ATP by inhibiting expression of unnecessary genes. When glucose is depleted and lactose is available, *lac* genes are again expressed to make use of available energy sources.

Time spent: 0 sec
Subject:
Biology

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System:
Cellular Biology

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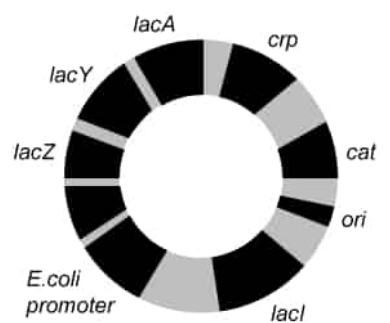


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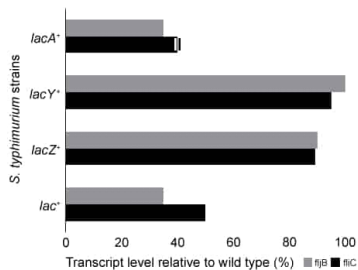


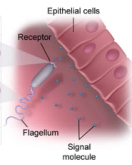
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Which of the following conclusions regarding virulence of *lac*⁺ *S. typhimurium* is most accurate?

- ☐ A. It cannot detect the presence of epithelial cells. [5%]
- ☐ B. It lacks the type III secretion proteins needed for invasion. [11%]
- ☐ C. It infects epithelial cells without using type III secretion proteins. [18%]
- ☒ D. It senses epithelial cells but cannot move toward them. [63%]

Explanation:

Genes	Fold change (<i>lac</i> ⁺ /WT)	p-value
Chemosensation	<i>cheZ</i> 0.87	0.620
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Chemotaxis is the process by which mobile cells both sense and move toward or away from an increasing concentration of a specific chemical. Many pathogens find host cells to invade by sensing the chemicals that they release and moving toward them. Impaired ability to either sense or respond to chemical signals can reduce virulence.

Table 1 illustrates the fold-change between WT and *lac*⁺ bacterial expression of several genes involved in invasion of host cells, calculated as the expression level measured in *lac*⁺ *S. typhimurium* divided by the expression level measured in WT *S. typhimurium*. A p-value indicates the probability that an observed change could have occurred due to random chance, with $p < 0.05$ generally being considered the threshold for a change to be considered statistically significant.

Based on the information in the table, the only genes with a **statistically significant decrease in expression level** ($p < 0.05$) are the **flagellar genes** (*fliB* and *fliC*) that code for proteins in the flagellum of *S. typhimurium*. The chemosensation genes are not significantly affected by the presence of the *lac* genes, so the bacteria are likely still able to detect the chemical signals produced by epithelial cells. However, impaired flagella would inhibit motion toward the signal's source.

(Choice A) The transcription of chemosensation genes is not significantly affected by the *lac* operon ($p > 0.05$), so *S. typhimurium* bacteria are most likely able to detect epithelial cells adequately.

(Choices B and C) *lac*⁺ *S. typhimurium* bacteria show no significant change in the expression of type III secretion genes or protein levels.

Educational objective:

Bacteria respond to chemical stimuli in the extracellular environment by moving away from or toward increasing concentrations of signal molecules. This movement, known as chemotaxis, allows bacteria to adjust the direction of their movement toward target cells and is an important part of infection. Impairing the ability to detect or respond to chemical signals can reduce virulence.

Time spent:0 sec
Subject:
Biology

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System:
Cellular Biology

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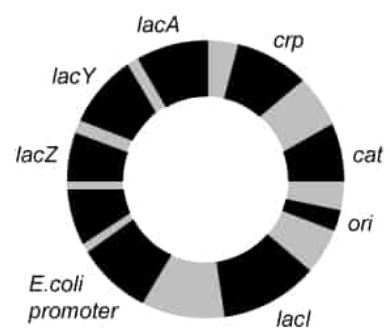


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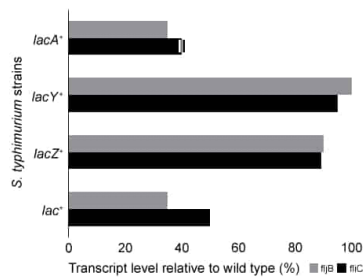


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Which table shows the expected survival rates of mice infected with WT or *lac*⁺ *S. typhimurium* after being fed a glucose-rich (GR) or a lactose-rich (LR) diet for several weeks?

☐ A.

Genotype	Diet	Mice Survival Rate (%)
WT	GR	40
	LR	0
<i>lac</i> ⁺	GR	30
	LR	61

[20%]

✓ ☒ B.

Genotype	Diet	Mice Survival Rate (%)
WT	GR	0
	LR	30
<i>lac</i> ⁺	GR	45
	LR	60

[40%]

☐ C.

Genotype	Diet	Mice Survival Rate (%)
WT	GR	55
	LR	20
<i>lac</i> ⁺	GR	70
	LR	0

[20%]

☐ D.

Genotype	Diet	Mice Survival Rate (%)
WT	GR	25
	LR	50
<i>lac</i> ⁺	GR	15
	LR	40

[18%]

Correct

39% answered correctly

Explanation:

	Genotype	Diet	Mice survival rate (%)	
Virulent	WT	GR	0	Cannot replicate well
		LR	30	
Less virulent	<i>lac</i> ⁺	GR	45	Inefficient infection
		LR	60	

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Different bacterial behaviors, such as virulence, can be induced or repressed by the surrounding environment. To be virulent, *Salmonella* strains must be able to enter cells and replicate. According to the passage, the ***lac* operon reduces virulence**.

Therefore, when measuring the survival rates of **infected mice**, mice infected with *lac*⁺ bacteria will have a **higher survival rate** than mice with the same diet but infected by WT bacteria.

The passage also says that **virulence is increased in the presence of glucose**. For WT bacteria, this is likely because they cannot digest lactose and therefore cannot replicate efficiently when lactose is the primary source of energy. In the case of *lac*⁺ bacteria, because lactose induces increased transcription of the *lac* operon, virulence is expected to decrease in lactose-rich conditions. Therefore, mice that were fed a glucose-rich diet would be expected to have lower survival rates than those fed a lactose-rich diet, regardless of the genotype of the invading bacteria.

(Choice A) *lac*⁺ bacteria are less virulent than WT, so *lac*⁺-infected mice should have a *higher* survival rate. However, this Answer Choice shows a *lower* survival rate when fed a glucose-rich diet.

(Choice C) The bacteria are *more virulent in glucose-rich conditions* than in lactose-rich conditions, so the mouse survival rates for a given strain should be *lower*, not higher, when glucose is abundant.

(Choice D) *lac*⁺ bacteria are less virulent than WT, so *lac*⁺ infection should have a *higher* mouse survival rate. However, this choice shows a *lower* survival rate for mice given either a glucose- or lactose-rich diet.

Educational objective:

Bacterial virulence can be induced or repressed by the surrounding conditions. The *lac* operon makes *lac*⁺ bacteria less virulent than the wild-type. In the presence of lactose, both strains become less virulent because they either cannot survive (WT) or cannot infect cells (*lac*⁺).

Time spent: 0 sec

Subject:
Biology

QID: 400483

System:
Cellular Biology

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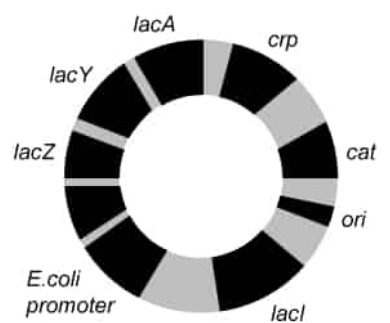


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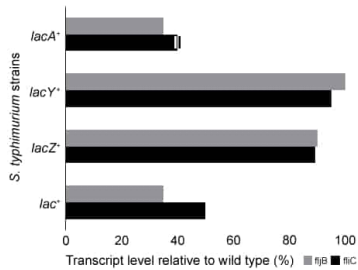
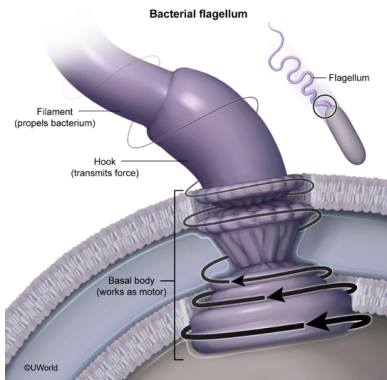


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The reduced virulence of *lac*⁺ *S. typhimurium* has been correlated with an inability to generate the torque that turns the flagellum. Which structure in the flagellum is most likely being affected by *lac* genes?

- ☐ A. Hook [32%]
- ☐ B. Filament [24%]
- ☒ C. Basal body [41%]
- ☐ D. Cell membrane [1%]

Explanation:

The virulence of *S. typhimurium* decreases due to impaired generation of flagellar rotation. Flagella in prokaryotes consist of three basic parts: a basal body, a hook, and a long helical tube called a filament. The component that generates torque is the **basal body**, a transmembrane unit that works as a **molecular motor**. Cells that are unable to generate torque likely have a defect in flagellar basal bodies.

(Choice A) The hook is a flexible structure that connects the basal body and the filament. It transmits torque generated by the basal body but does not generate any rotation itself.

(Choice B) Filaments, which are mostly composed of flagellin, are the recipients of the rotation generated by the basal body. They respond to rotation and propel bacteria but do not generate rotational motion.

(Choice D) The cell membrane helps hold the flagellum in place but does not generate flagellar motion.

Educational objective:

Bacterial flagella are composed of a basal body, a hook, and a hollow filament filled with flagellin. The hook connects the filament to the basal body, and the basal body serves as the motor that generates motion.

Time spent: 0 sec
Subject:
 Biology

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System:
 Cellular Biology

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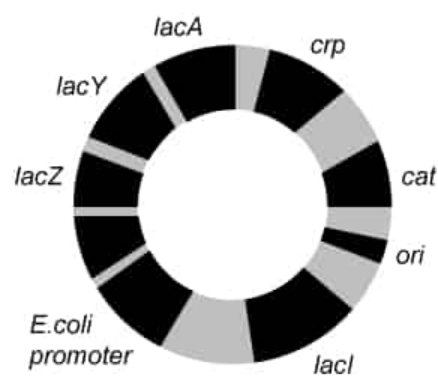


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Type III secretion	<i>invF</i>	0.76	0.730
	<i>invI</i>	0.95	0.920

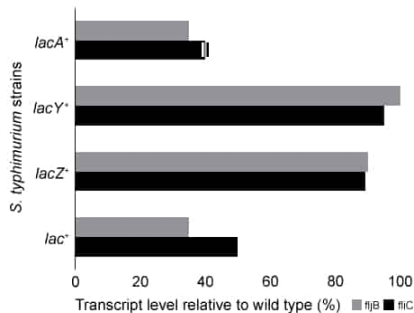
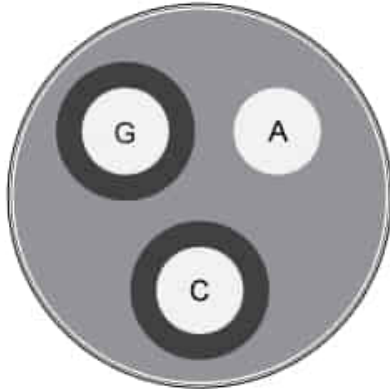


Figure 2 Transcript levels of two *S. typhimurium* flagellar genes, *fliB* and *fliC*, in bacteria transformed with *lacZ*, *lacY*, *lacA*, or *lac* plasmids. Levels are portrayed as a percentage of WT levels.

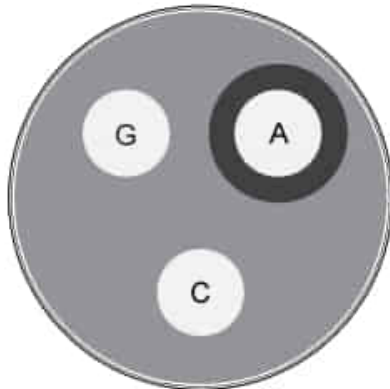
Three antibiotic disks were placed on a bacterial culture. One disk contained gentamicin (G), one contained chloramphenicol (C), and one contained ampicillin (A). Based on this information, which of the following cultures represents a strain of Salmonella that has been transformed with the plasmid shown in Figure 1?

☐ A.



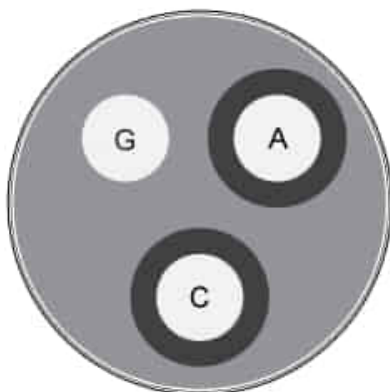
[16%]

☐ B.



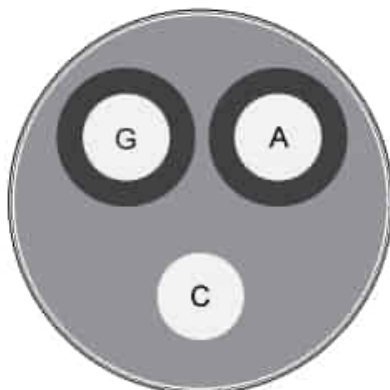
[9%]

☐ C.

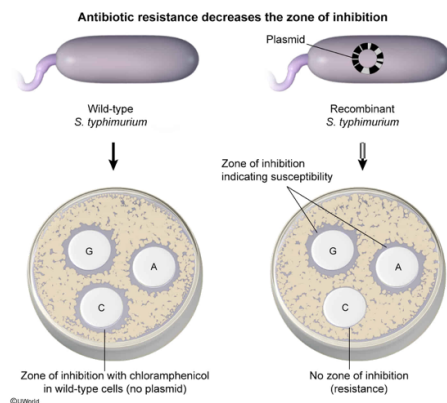


[12%]

✓ ☒ D.



[61%]

Explanation:

Antibiotic resistance is often investigated using the agar diffusion test in which disks impregnated with different antibiotics are placed on a bacterial culture. **Susceptible bacteria** normally form a growth-free halo called a **zone of inhibition** around the disks due to the high antibiotic concentrations. However, **resistant strains can grow** near that antibiotic and *do not* form a zone of inhibition.

Plasmid DNA used to transform bacteria with a gene of interest (*lac*, in this case) generally also includes an antibiotic resistance gene so that untransformed bacteria can be selectively eliminated. The passage indicates that plasmids containing *lac* genes also have **chloramphenicol resistance** genes. Therefore, **transformed bacteria** should be able to **grow near the chloramphenicol disk** but not the other antibiotics.

(Choices A, B, and C) *lac*⁺ cells will not form a zone of inhibition around chloramphenicol because the transforming plasmid contains a chloramphenicol-resistance gene. However, they *will* form zones of inhibition around ampicillin and gentamycin.

Educational objective:

Antibiotic resistance genes in plasmids are used to select bacteria with a gene of interest. In an agar diffusion test, bacteria that are susceptible to an antibiotic will form a zone of inhibition around the antibiotic disk where they cannot grow. Bacteria with resistance genes can grow close to the disk.

Time spent: 0 sec

Subject:
Biology

QID: 400485

System:
Cellular Biology